From a few to the whole crowd: a practical introduction to sampling and survey data analysis.

Workshop (2 h)

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Summary

Obtaining information about a large population by selecting and measuring a sample from that population is common epidemiological practice and, in this context, the use of complex sampling strategies is ubiquitous.

However, the use of these strategies – such as clustering and stratification – has implications in terms of how the data should be analysed and how the population estimates are to be interpreted. A deep understanding of the links between sampling strategy and realisation, sample sizes, analysis techniques and accuracy of the population estimates is essential for optimal study planning and, similarly, for secondary analysis of data previously collected.

During this workshop, we will conduct a series of 'virtual surveys' on a simulated human population of approximately 40 000 individuals inhabiting a hypothetical territory remotely accessible through a graphical web interface (the *SurveyLab*). We will discuss the results of these experiments as 'case studies' to get a practical understanding of the pros and cons of alternative sampling strategies and statistical methods and how poor choices can lead to profoundly misleading results and/or waste of resources. We will use the data gathered during the virtual surveys to illustrate alternative approaches that can be used to analyse this kind of information.

Prerequisites

Previous knowledge or experience with survey data analysis is not required, but participants are assumed to be familiar with basic epidemiological and statistical concepts (such as bias, precision and probability sampling). A (short) recap of the statistical foundations of the techniques applied will introduce the workshop.

Data analysis examples will be presented using R environment for statistical computing (https://www.R-project.org/).

Target

Researchers, students, and professionals who want to acquire a deeper understanding of issues related to the analysis and interpretation of data gathered from samples of large populations.